

Qian Chen

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EDUCATION

Zhejiang University–UIUC Institute, Undergraduate Student Sep. 2023–Jun. 2027 (Expected)

Zhejiang University *B.S. in Mechanical Engineering* GPA: 3.84/4.0; Rank: 6/56
University of Illinois Urbana-Champaign *B.S. in Mechanical Engineering* GPA: 4.0/4.0

Core Coursework: Mechanical System Dynamics, Solid Mechanics, Engineering Materials, Thermodynamics, Fluid Mechanics, Mechanical Design, Signal Processing, Analog Circuits and Systems, Statistics and Probability

Research Interests: Robotics, Embodied Intelligence, and Automatic Control

PROFILE

Mechanical engineering undergraduate with strong academic performance, self-motivation, teamwork, and English communication skills. Through projects in robot mechanical design and reinforcement-learning-based motion control, I have developed integrated capabilities spanning mechanism implementation, simulation modeling, and experimental validation. In graduate study, I hope to focus on embodied-intelligence robotics and automatic control, exploring perception, decision-making, and motion control for robots operating in complex environments.

RESEARCH AND PROJECT EXPERIENCE

Biped Locomotion Manipulation

Jun. 2026–Present
Advisors: Hua Chen and Liangjing Yang, ZJUI

- Conducting preliminary research and experimental setup for deploying the Umi system on legged robot platforms, with emphasis on coordination between mobile platforms and manipulation tasks.
- Continuing system design and integration while building research experience in legged robotics and embodied manipulation.

Reinforcement Learning for Omni-Wheel Biped Locomotion on Complex Terrain

Jan. 2026–Jun. 2026
Advisor: Hua Chen, ZJUI

- Studied the codebase, training workflow, and control framework for reinforcement-learning-based locomotion; configured IsaacLab and MuJoCo Playground, reproduced PPO policies, and analyzed robot models, observations, reward design, and deployment results.
- Completed the full workflow from simulation training and sim-to-sim validation to sim-to-real deployment, establishing a foundation for subsequent robotics research.

Mechanical Design and Optimization for RoboMaster Competition Robots

Apr. 2024–Apr. 2026
Advisor: Jiahuan Cui, ZJUI

- Contributed to robot mechanical design for chassis stability, projectile-feeding reliability, and power-module protection; used SolidWorks and Fusion 360 to redesign suspension connectors, the magazine, feeding dial, and supercapacitor protection structures, followed by fabrication and assembly validation using 3D printing, CNC machining, and laser cutting.
- As a principal team lead, led the team in the RoboMaster University League and earned Second Prize in the regional competition.

Automated Micro-Feeding Device for Timed and Quantitative Single-Fly Feeding

Feb. 2026–Apr. 2026
Advisors: Timothy Haw-Yu Lee and Jiahuan Cui, ZJUI

- Designed CAD models, mechanical structures, and fabrication processes for timed and quantitative micro-feeding in single-fly culture; optimized moving-component layout and actuation to improve space efficiency and ease of assembly.
- Completed fabrication and assembly of the automated single-fly micro-feeding device to support subsequent stable feeding experiments.

**Magnetic-Field-Assisted 3D Printing and
Optimization of Grooved Flexible Actuators**

Jun. 2024–Aug. 2024
Advisors: Peng Zhao and Chengqian Zhang, ZJU ME

- Used SolidWorks to design groove-parameter experiments and COMSOL to support magnetic-field simulation; programmed an Arduino Mega controller and combined stereolithography 3D printing with a magnetic-field setup to validate soft-robot motion.
- Completed the summer research project and produced a technical research paper.

HONORS

Second Prize and Third Prizes, RoboMaster University League Shanghai Regional (Principal Team Lead)	2024–2026
Scholarships: Second and Third Prizes, Zhejiang University; Third Prize, ZJU-UIUC Institute	2024, 2025
Third Prize, 5th Zhejiang University Student Concrete Dragon Boat Competition (Captain)	2024
S Award, Mathematical Contest in Modeling	2025
Dean’s List of Grainger Engineering, UIUC	Fall 2025

TEACHING ASSISTANT EXPERIENCE

Teaching Assistant, Zhejiang University	Jan. 2026–Jun. 2026
<i>Introduction to Solid Mechanics</i>	Instructor: Ye Shi
<i>Introductory Fluid Mechanics</i>	Instructor: Jiahuan Cui

SKILLS

Hardware: Fusion 360, SolidWorks, COMSOL, and 3D printing
Programming: Python, MATLAB, Arduino, and LaTeX
Languages: CET-6: 546; TOEFL: 5.5/6